FUTURE INTERNS

TASK 2 PROJECT REPORT

CLIENT LEAD MANAGEMENT SYSTEM (MINI CRM)

Project name: LeadDesk
Internship track: Full Stack Web Development
Required GitHub repository: FUTURE_FS_02

1. Introduction

This project is a simple customer relationship management (CRM) application designed to help an administrator organise potential clients generated through website contact forms. It keeps lead information in one place and supports consistent follow-up activities.

2. Objectives

- Store and display client lead information.
- Track each lead using clear statuses.
- Record notes and schedule follow-up dates.
- Provide secure administrator access.
- Demonstrate CRUD operations, backend integration and database management.

3. Technologies Used

Frontend: React, Vite, JavaScript and CSS. Backend: Node.js and Express REST API. Database: MongoDB with Mongoose. Authentication and security: JSON Web Tokens, bcryptjs password hashing, Helmet security headers and login rate limiting.

4. Main System Features

- Administrator sign-in using credentials configured in the server environment.
- Dashboard summary showing total leads and counts for New, Contacted and Converted.
- Create, view, edit and delete client lead records.
- Update lead status to New, Contacted or Converted.

- Add timestamped notes to a client record and set a follow-up date.
- Search by client name, email or lead source and filter by status.
- Responsive interface for desktop and mobile screen sizes.

5. Database Design

The Lead collection stores the client name, email address, lead source, status, optional follow-up date, notes and creation/update timestamps. Each note stores its text and creation time. The User collection stores the administrator email and a securely hashed password. MongoDB provides persistent storage between sessions.

6. Security and Validation

Lead-management API routes require a valid JWT. Administrator passwords are hashed with bcryptjs. Helmet adds common HTTP security headers and login attempts are rate-limited. The API validates required fields, email format, permitted status values, follow-up dates and note length. Secret values are stored in a local environment file and excluded from Git.

7. Installation and Execution

The project package includes a README with the full setup procedure. MongoDB must be running or an Atlas connection must be configured. Install dependencies separately in the server and client folders, configure server/.env using server/.env.example, start the API, then start the Vite frontend. Open the local Vite address and sign in with the administrator credentials configured in the environment file.

8. Conclusion

LeadDesk demonstrates a full-stack workflow for managing client leads. It addresses the core Task 2 requirements by combining lead listing, status updates, notes and follow-ups, administrator authentication, backend APIs and persistent database storage.

9. Evidence for Final Submission

After running the application, capture your own screenshots of the sign-in screen, dashboard and lead list, add/edit lead form, and a client profile showing notes and follow-up information. Add the screenshots to the public GitHub repository if required by the evaluator.

Submission reminder: Create a PUBLIC GitHub repository named exactly FUTURE_FS_02 and upload the project source. Submit the repository link through the official Future Interns Task Submission Portal when it is provided. The application has not been deployed or uploaded to your GitHub account by this report.